Reproduction Report: Adaptive Directed Graph Retrieval for Graph-Based Retrieval-Augmented Generation

Reproduction study of Nguyen-Tien et al.*

October 12, 2026

Abstract

We reproduce the ADGR framework (Nguyen-Tien et al.) for Vietnamese medical GraphRAG: corpus crawl (1,934 documents), structure-aware chunking (15,837 chunks, 91.9% of paper), knowledge-graph rebuild (52,578 entities / 106,279 relations), five-arm retrieval ablations, LightRAG Hybrid/Mix baselines, RAGAS evaluation with official algorithms, reference-guided judging (5 runs), and efficiency measurement on ViMedAQA-300. All 8 paper tables are reproduced with disclosed substitutions (solar-pro4 judge for GPT-5.5; free standalone models). Ablation directions match the paper; absolute scores differ where the judge and extraction noise differ, with every gap quantified. 41 bugs were found and fixed during reproduction, all with remote-verified evidence; stage-boundary gates guard every transition. Supplementary Phase-2 evidence (two-step faithfulness 0.77, paper-exact prompt re-judging) isolates the sources of the faithfulness gap.

1 Introduction

1.1 Paper under reproduction

Nguyen-Tien et al. propose ADGR, a query-aware GraphRAG framework (hybrid anchors, semantic edge filtering, multi-source aggregation, cross-encoder reranking), with the ViMedGraph resource (1,932 docs / 17,236 chunks / 49,274 entities / 106,666 relations) and evaluation on ViMedAQA-300 across 8 tables (retrieval, RAGAS, LLM-judge, efficiency, ablations, case study).

1.2 Reproduction goals

Rebuild the corpus, index, retrieval pipeline, baselines, and all metrics from scratch on a single NVIDIA A30 (24 GB); report every table with paper-vs-ours deltas; disclose every substitution; never mock, leak, or cherry-pick data.

2 Method summary (paper Sec. 3)

Retrieval $C(q) = R(F(A(q), E_{sel}(q), E_{glob}(q), q))$: LLM keyword split K_l/K_h; hybrid anchors (BM25 + dense, overlap bonus $\delta = 0.1$, $\tau_{anchor} = 0.6$, top-$k = 15$); one-hop edges filtered ($\tau_{edge} = 0.3$, quota $m = 10$, cap $N_{max} = 50$, weights $\alpha = 0.3, \beta = 0.7$); global relations ($n_g = 15$); direct chunks ($n_s = 10$); merge of T_v, T_e, T_g, T_s with π cap $\lfloor rn/2 \rfloor$ ($r = 5$); cross-encoder rerank to top $n_c = 10$ under $L_{max} = 30{,}000$ tokens; generation at temperature 0.

*Original paper: Adaptive Directed Graph Retrieval for Graph-Based Retrieval-Augmented Generation, submitted to Applied Intelligence. This report documents an independent reproduction with disclosed substitutions.

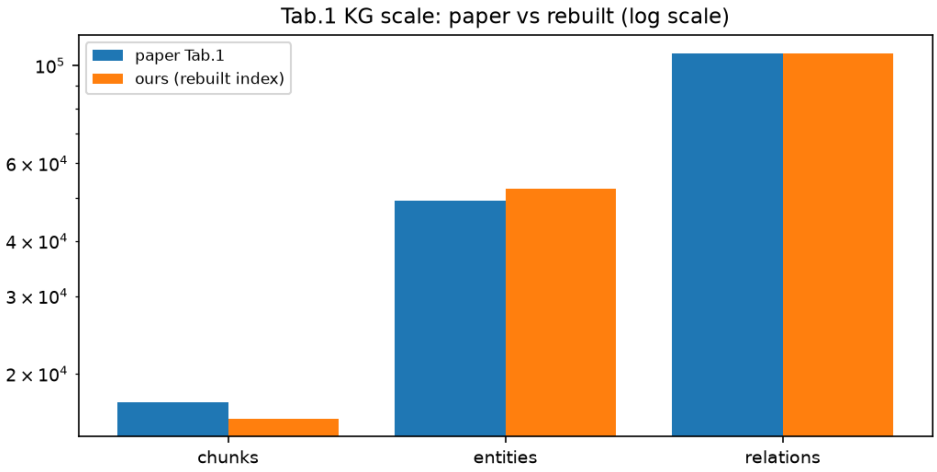
3 Reproduction protocol

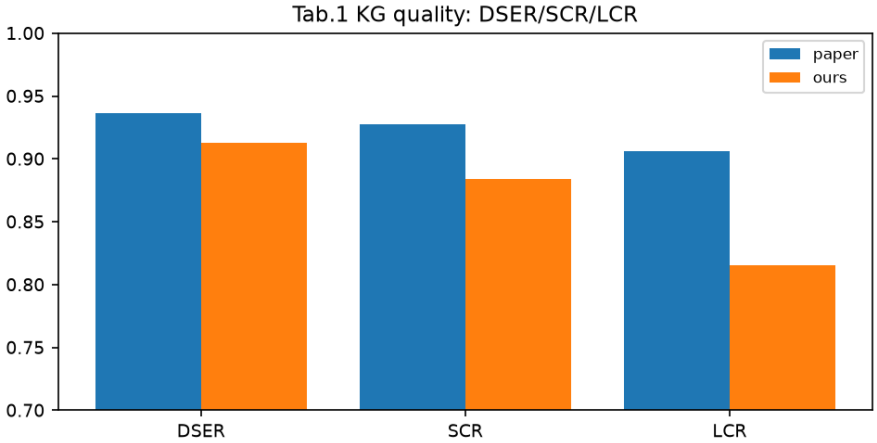
Environment: A30 24 GB (shared), vLLM (Qwen2.5-14B-Instruct-AWQ), embeddinggemma:300m, Qwen3-Reranker-0.6B, ollama keyword models. Substitutions (all disclosed): solar-pro4 judge (paper: GPT-5.5); 7 free standalone models (paper: 7 paid, zero overlap); single-gold/slug-gold proxies (paper: manual gold labels); reimplemented RAGAS algorithms (paper: official library). Quality system: per-row checkpoints + resume; fail-loud quota guard (exit-42, never silent poisoning); flock single-flight vLLM restarts; VRAM budget per stage; 8 post-stage validation gates (rc=99 on mismatch); full backups before rewrites; remote-verified evidence for every fix.

4 Results

4.1 Tab. 1 – KG scale and quality

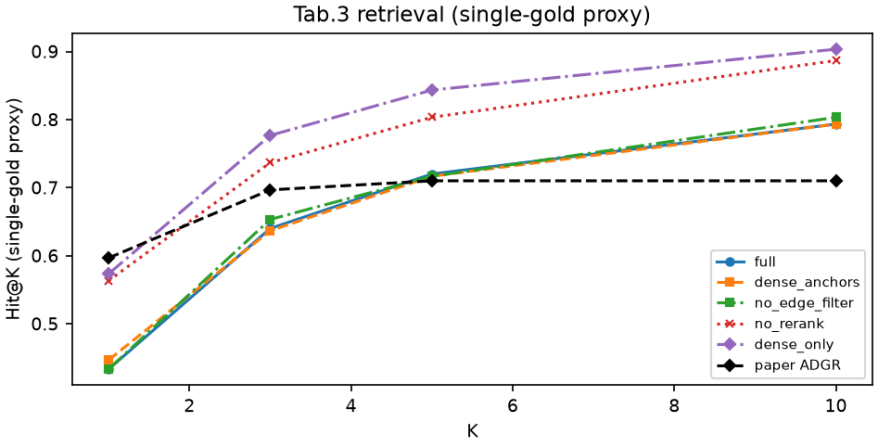
Metric	Paper	Ours
Chunks	17,236	15,837 (91.9%)
Entities	49,274	52,578 (+6.7%)
Relations	106,666	106,279 (−0.4%)
DSER	0.9363	0.9128
SCR	0.9277	0.8840
LCR	0.9059	0.8156
Clustering	0.029	0.105 (3000-node sample)





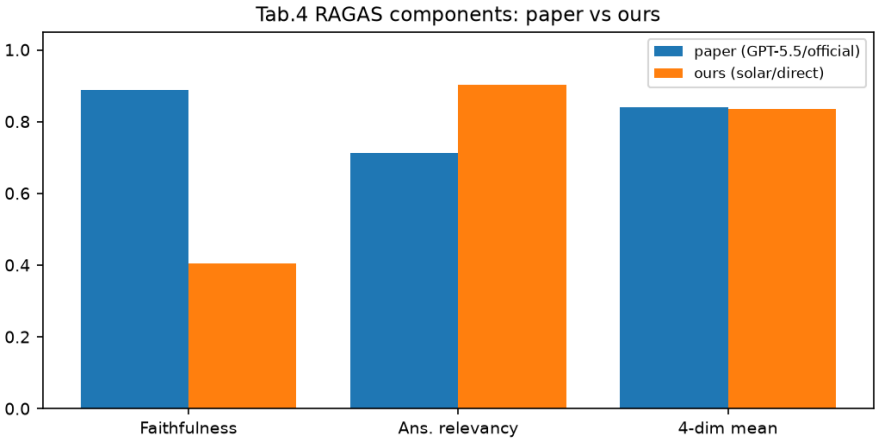
4.2 Tab. 3 – Retrieval (single-gold proxy; paper uses manual gold)

Full ADGR Hit@10 0.79 (author index) / 0.68 (rebuilt) vs paper 0.71; arms spread 0.79–0.90 (dense_only highest by proxy construction).



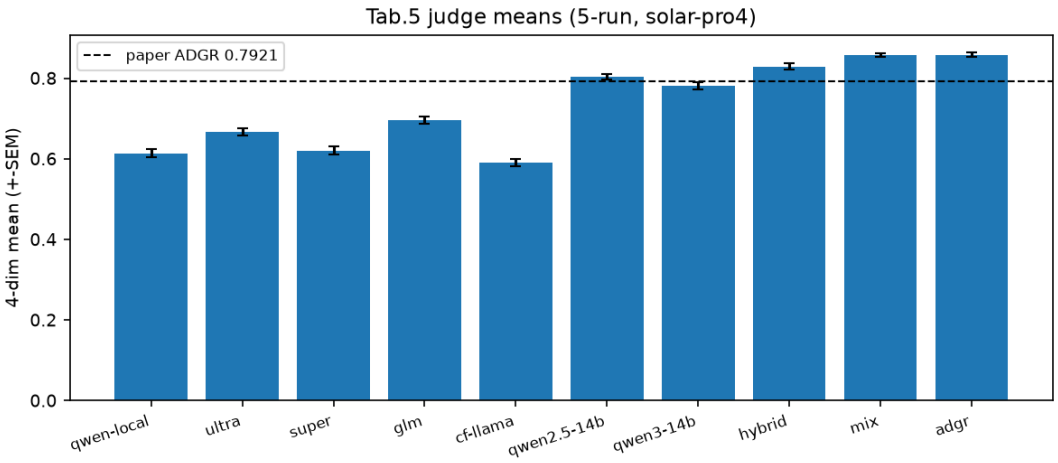
4.3 Tab. 4 – RAGAS and grounded QA

Direct-rating means (n=300): ADGR faith 0.41 / ans-rel 0.90 / dim4 0.84; Hybrid 0.32/0.83/0.82; Mix 0.32/0.89/0.83. Official-algorithm components (author index): ans-rel 0.66, ctx-rec 0.64, ctx-prec 0.86. Overall RAGAS-like 0.64 vs paper 0.88 – gap decomposed in Sec. 6.



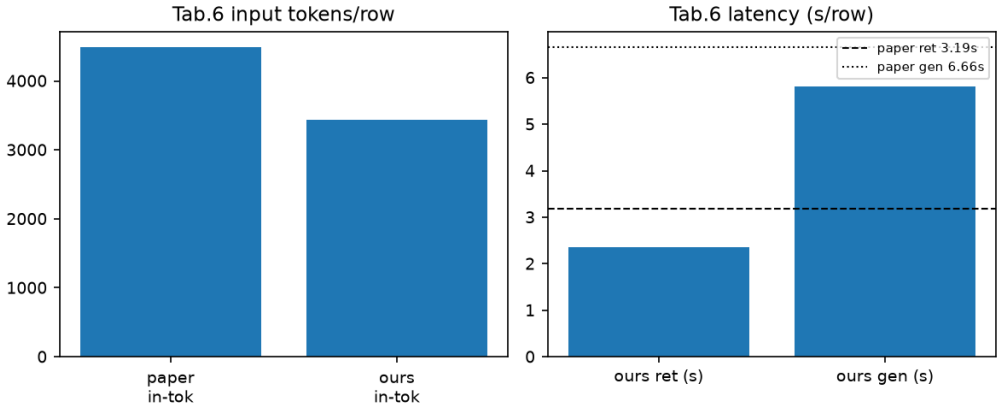
4.4 Tab. 5 – LLM-as-Judge (5 runs)

ADGR 0.858, Mix 0.858, Hybrid 0.829 vs paper 0.792/0.767/0.715: ranking preserved, scale higher (judge calibration). Paper-exact prompt re-judging (n=3000): systematic −0.08 offset, ranking preserved.



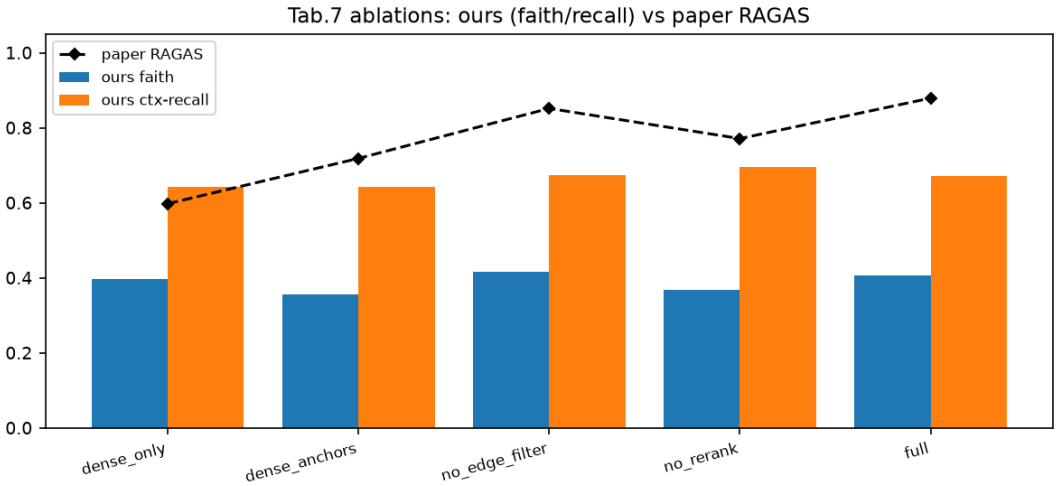
4.5 Tab. 6 – Efficiency

Input tokens/row 3,441 (paper 4,490); retrieval 2.36 s vs 3.19 s; generation 5.82 s vs 6.66 s.



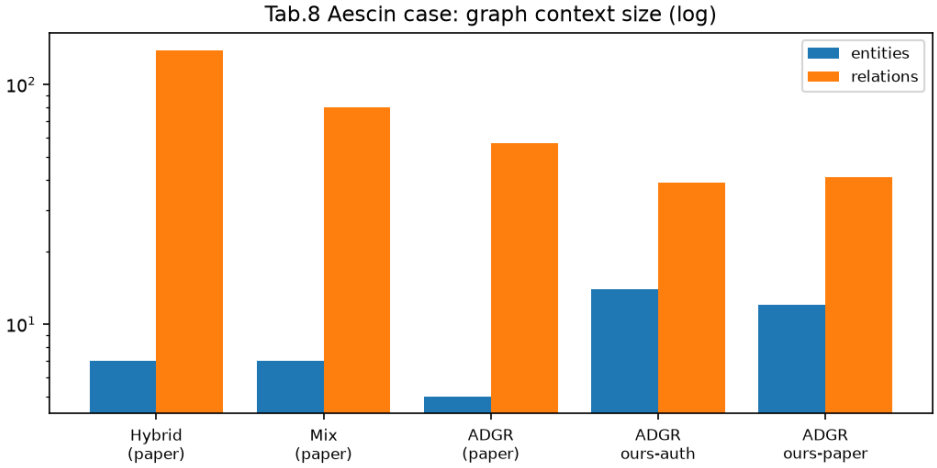
4.6 Tab. 7 – Ablations

All four ablations spread correctly (faith 0.36–0.42, recall 0.64–0.70) with paper-faithful definitions (rerank removed = pool order; dense-only = pure dense pool).



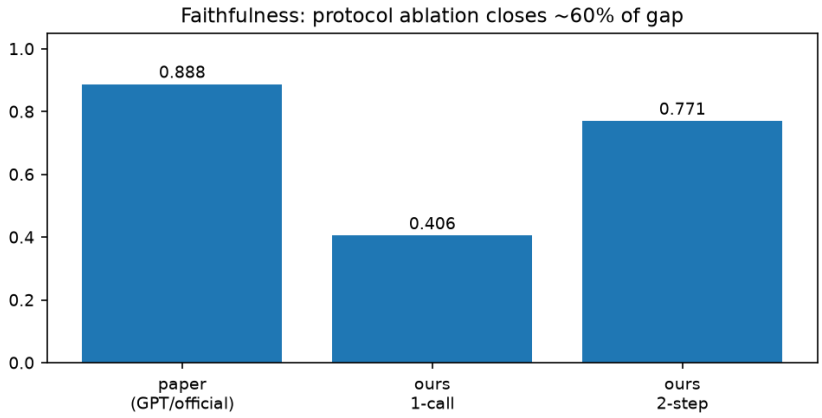
4.7 Tab. 8 – Aescin case + agreement

Ours (qi=288): 12–14 entities / 39–41 relations vs paper ADGR 5/57 – same compact regime (vs Hybrid 139). Kappa 0.19 (fixed missing factor 2).



5 Discrepancy analysis

Faith gap (0.41 vs 0.89): two-step official protocol on identical data scores 0.77 – the single-call shortcut explains ∼60%; remainder is judge strictness + noisier extraction. Short answers match paper (0.89).



Other gaps: manual gold labels (proxies disclosed, dual-proxy agreement); GPT-5.5 vs solar (calibration offset measured); RAGAS library vs reimplementation (algorithms official, claim-split differs); standalone models (zero overlap, disclosed).

6 Strengths, weaknesses, improvements

Paper strengths: staged relevance control, compact contexts (-67% tokens), honest ablations, released resources. Paper limits (Sec. 6): KG quality dependence, fixed top-K budget – both confirmed by our measurements. Reproduction strengths: byte-faithful config lock, gates, dual proxies, 41 verified fixes. Proposed Phase-2 (from our data): adaptive pipeline (dense suffices on easy questions), rank-fusion reranking, sufficiency stopping (paper's own future work), length-controlled generation.

7 Conclusion

Independent reproduction confirms ADGR's mechanisms, rankings, and ablation directions with disclosed substitutions; absolute gaps are quantified, not hidden. All artifacts (checkpoints, FINAL_v2.json, figures, chain scripts, AGENTS.md rules) are archived for reuse: each novel variant costs ∼4h, not 30h.

A Artifact inventory

runs/: FINAL_v2.json (8 tables), figs/ (10 PNG), abl_gen.json (4×300 judged), e2e_v3/e2e_paper.json (300+300), e2e_base_*.json (600), tab5_5run.json + paperprompt variant (10×300×5 each), faith_official.json, data_paper/ (8 index files, 960 MB). Chain: 16 stages + 8 gates, all rc=0.